

Six Sigma and the 6 M's

The primary objective of the six sigma methodology is to reduce variation.

The 6 M's are the six major sources of variation.

Machine

Method

Man

Material

Measurement

Mother Nature

The 6 M's serve as a framework for brainstorming potential root causes.

Names like "Man" and "Mother Nature" could easily be replaced with "Employee" and "Ambient Conditions".



Six Sigma and the 6 M's

Machine

Machinery

Control Software

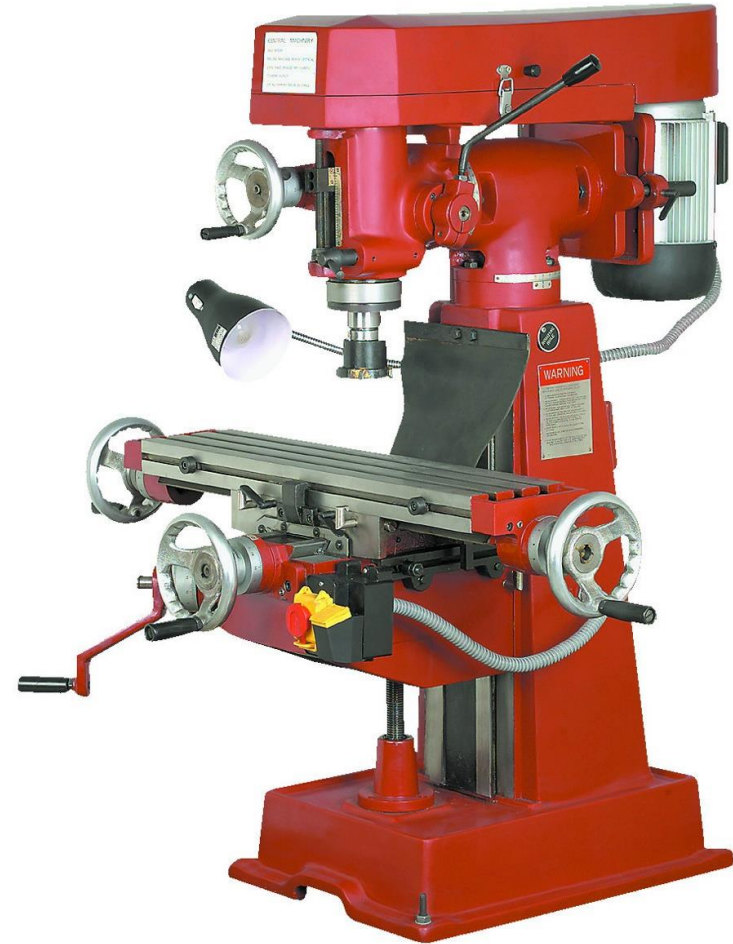
Perishable Tooling

Fixturing / Jigs

Corrective / Preventive Maintenance

Facilities

Includes capacity and capability of the machinery.



Six Sigma and the 6 M's

Method

Feeds / Speeds

Process "Recipe"

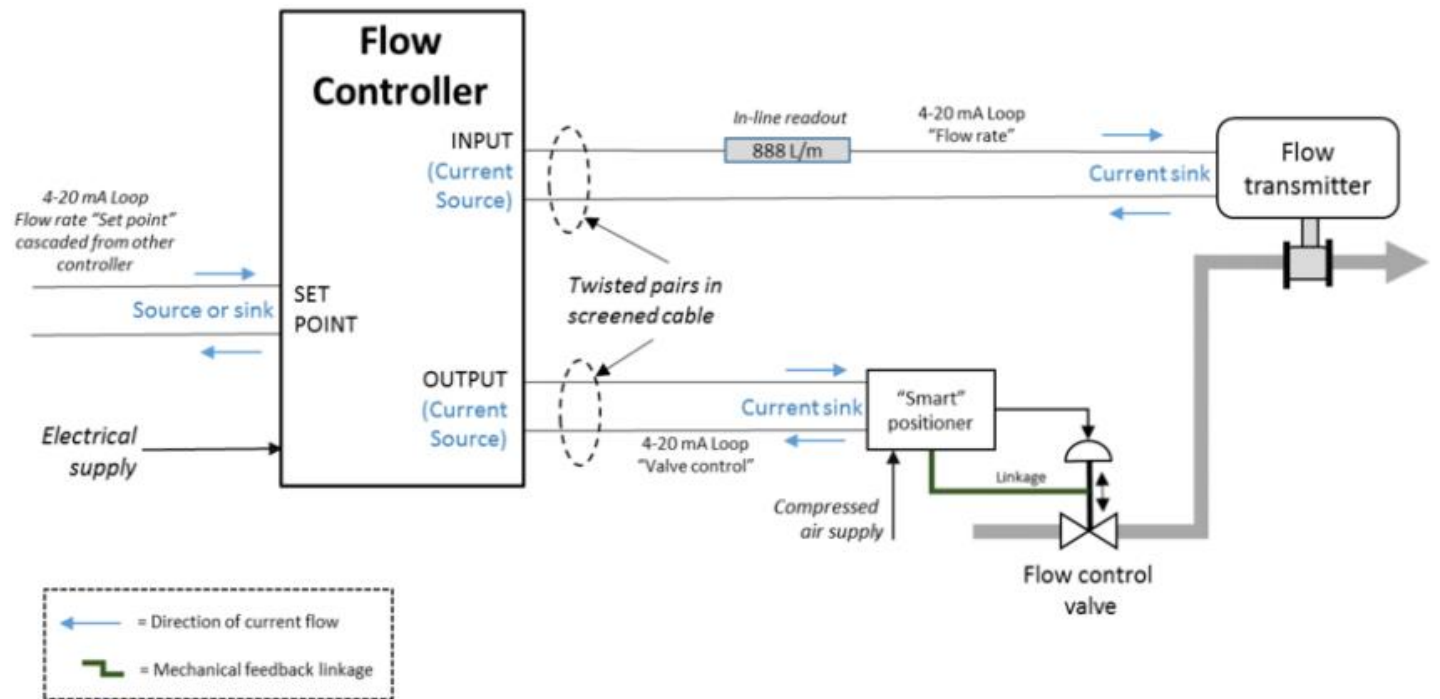
Sequence of operations

Work instructions

Visual aids

Error proofing devices

Signaling devices



Six Sigma and the 6 M's

Man

(The “People” factor)

Training

Experience

Empowerment

Habits

Sense of Responsibility

Technical Proficiency

Awareness

Physical strength / Fatigue / Ergonomics

Includes total available labor



Howard R. Hollem, “A factory worker in 1940s Fort Worth, Texas”

Six Sigma and the 6 M's

Materials

Chemistry / Physical Properties

Materials Packaging

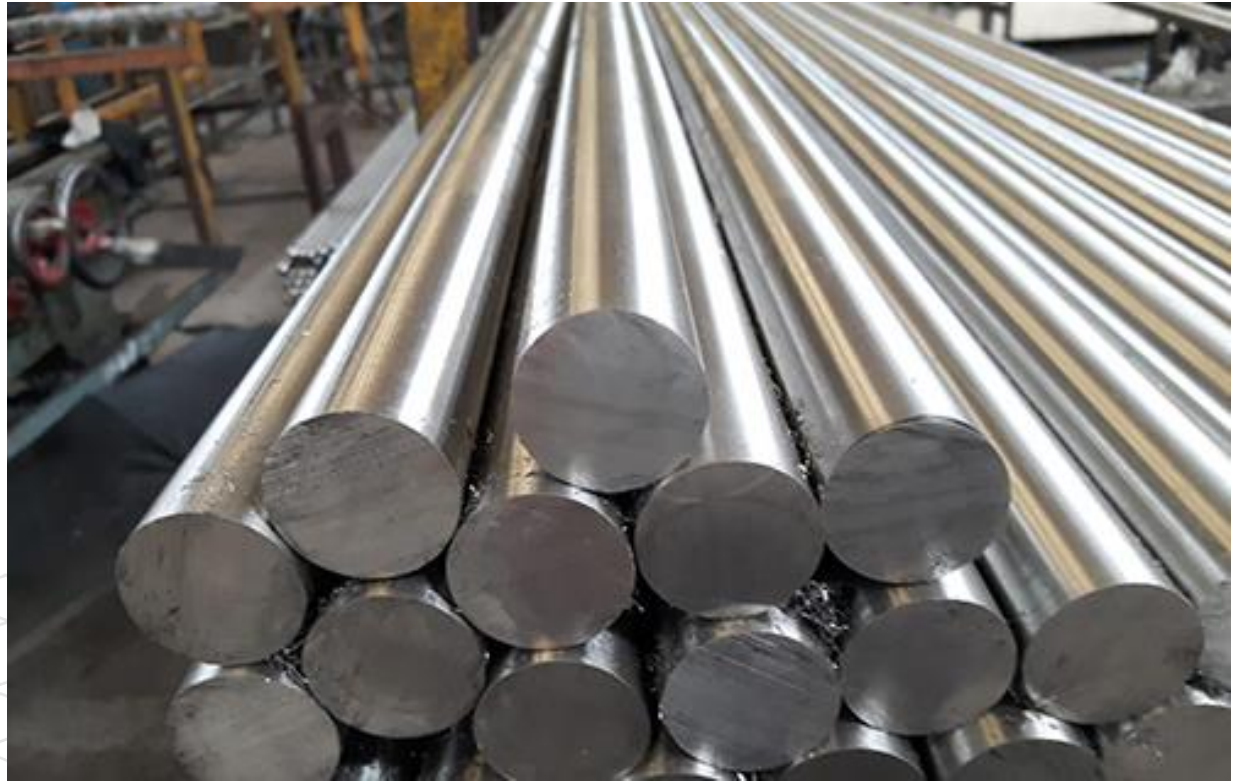
Quality of Purchased Components

Consumables (Grease, Oils,
Coolants, etc.)

Labeling

Storage

Perishability



Six Sigma and the 6 M's

Measurement

Gage Types and Design
Measurement Frequency
Sampling Size
Bias, Stability and Linearity
Data Collection Systems
Measurement Records



Six Sigma and the 6 M's

Mother Nature (Ambient Conditions)

Temperature

Humidity

Dust

Vibration

Noise

Light

